

A new path-following constraint for strain-softening finite element simulations

E. Lorentz^{1,*},[†] and P. Badel^{1,2}

¹*Electricité de France - R&D, 1, av. Général de Gaulle, 92141 Clamart cedex, France*

²*Laboratoire de Modélisation en Mécanique, Université Paris 6, Place Jussieu, 75005 Paris, France*

SUMMARY

The application of strain-softening constitutive relations to model the failure modes of real-life structures is faced to numerical difficulties related to instabilities that appear as sharp snap-backs of the structural response. A path-following method has to complement the solution algorithm to achieve convergence despite these critical points. Because of the sharpness of the snap-backs, it is believed essential that the path-following constraint distinguish between a purely elastic unloading and a dissipative path. For that purpose, a new constraint based on the maximal value of the elastic predictor for the yield function is proposed. As it is highly non linear, a specific solution algorithm is required. The robustness of this constraint is illustrated by three applications: the study of crack propagations by means of a cohesive zone model, the failure of a structure submitted to nonlocal damage and the simulation of a nonlocal strain-softening plastic specimen. Copyright © 2004 John Wiley & Sons, Ltd.

KEY WORDS: strain-softening; damage; instability; path-following method; finite element

1. INTRODUCTION

The prediction of the mechanical failure of real-life structures is of paramount industrial interest. An answer to this challenge may be given by the local approach to fracture where physical degradations are modelled at the material point scale through appropriate constitutive relations (applications to metal, concrete, soil, etc.). They share a characteristic feature, strain-softening, which often results in unstable structural responses [1, 2] related to the brutal growth of damaged zones. These instabilities are generally caused by an elastic energy release (elastic unloading) due to stress redistributions higher than the dissipation of the damage process [3]. They appear as snap-backs on a force–displacement curve, that is a simultaneous decrease of the force and the displacement.

On a numerical ground, snap-backs preclude the convergence of the iterative algorithm used to solve the non linear problem as soon as a monotonous force control or a monotonous dis-

*Correspondence to: E. Lorentz, Electricité de France, Department of Analyses in Mechanics and Acoustics, 1, av. Général de Gaulle, 92141 Clamart cedex, France.

[†]E-mail: eric.lorentz@edf.fr

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placement control are considered. In addition to these physical instabilities, spurious snap-backs also originate from the spatial discretization [4, 5] and strengthen still more the convergence difficulties. Passing these critical points has been the subject of intensive research in the past decades, which result in various path-following methods.

The first one is the direct displacement method [6, 7] where the load level is controlled through the evolution of a single degree of freedom. It implicitly assumes that the evolution of at least one degree of freedom is monotonous. However, such a degree of freedom does not always exist and, even in that case, its determination is not systematic. A straightforward enhancement of this method [8] consists in replacing the control of a single degree of freedom by a weighted combination of some or all of them but it may suffer from the same drawback. Another technique is the arc-length method, initially proposed in References [9, 10] then enhanced in References [11, 12]. The control of the load level is now achieved through a path-following constraint which prescribes either the (quadratic) norm of a combination of displacements and forces (spherical arc-length) or the norm of a set of displacements only (cylindrical arc-length). Although this approach has proved efficient for geometrical instabilities (buckling), it may fail to deal with simulations of material instabilities [8]. To enhance its robustness, a new criterion has been proposed in Reference [13] to select the best solution to the (quadratic) path-following constraint, but it will be shown that it may prove insufficient. Both classes of methods, direct displacement and arc-length, are encompassed within a unified framework proposed in Reference [14], where the control of the load level is achieved through several functions of part or all the components of the nodal displacement vector. Numerical evaluation of a significant number of control functions and strategies to select the active one have been analysed.

In spite of the large amount of work on the subject, we think that a question has not yet been addressed concerning brittle instabilities. On a force–displacement curve, the snap-backs are close to true elastic unloading. Therefore, the path-following constraint should explicitly distinguish between elastic and dissipative solutions to guaranty that only the latter be selected. We believe that this is a key feature to ensure the necessary robustness required by industrial applications. This article aims at proposing and evaluating such a constraint for rate-independent materials whose constitutive relations rely on a yield threshold. Its definition, based on the maximal value reached by the elastic predictor for the yield function, is given in Section 2 and a possible solution strategy is proposed in Section 3. Then, Sections 4–6 demonstrate the robustness of this constraint and the efficiency of the solution algorithm when applied to crack propagations (cohesive crack model), non-local damage and nonlocal strain-softening plasticity.

2. A PATH-FOLLOWING CONSTRAINT BASED ON THE MAXIMAL ELASTIC PREDICTOR FOR THE YIELD CRITERION

2.1. *Standard formulation*

Continuation methods applied to non-linear problems in mechanics consist in introducing a new scalar unknown that controls the level of a part of the load, while a new equation is added to the set of equilibrium equations, see for instance Reference [15]. After spatial discretization by finite elements and temporal discretization of the constitutive law, the non-linear system that

characterizes the mechanical solution for a given increment reads in a formal way

$$\begin{aligned}\mathbf{F}_{\text{int}}(\Delta\mathbf{U}) + \mathbf{B}^T\boldsymbol{\lambda} &= \mathbf{F}_{\text{ext}}^0(\mathbf{U}) + \eta\mathbf{F}_{\text{ext}}^1(\mathbf{U}) \\ \mathbf{B}\mathbf{U} &= \mathbf{U}_{\text{imp}}^0 + \eta\mathbf{U}_{\text{imp}}^1 \\ P(\Delta\mathbf{U}, \eta) &= \Delta\tau\end{aligned}\quad (1)$$

The unknowns of this set of equations are the displacement field $\mathbf{U}$ (or equivalently its increment $\Delta\mathbf{U}$), the vector of Lagrange multipliers $\boldsymbol{\lambda}$ that dualize the kinematic relations [16] and the load factor η . The first equation expresses the equilibrium: $\mathbf{F}_{\text{int}}$ denotes the (opposite of the) internal forces and depends on the displacement increment (and the set of internal variables evaluated at the beginning of the increment) through temporal integration of the constitutive law, $\mathbf{B}^T\boldsymbol{\lambda}$ represents the reaction forces and $\mathbf{F}_{\text{ext}}^0$ and $\mathbf{F}_{\text{ext}}^1$ are, respectively, the fixed and adjustable parts of the applied forces, which may depend on the displacements. The second equation expresses the kinematic relations (assumed to be linear in the present case), including Dirichlet conditions, through the matrix $\mathbf{B}$ and the vectors $\mathbf{U}_{\text{imp}}^0$ and $\mathbf{U}_{\text{imp}}^1$, the former being fixed while the latter is adjustable. The third equation is the path-following constraint: $\Delta\tau > 0$ is the step-length, while P is the newly introduced control function which governs the load factor η .

A way to solve this system based on the Newton–Raphson algorithm has been proposed in Reference [7]. $\mathbf{U}^n$, $\boldsymbol{\lambda}^n$ and η^n denote the unknowns at iteration n . A linearization of the two first equations of (1) with respect to $\mathbf{U}$ and $\boldsymbol{\lambda}$ leads to (with $\mathbf{U}^{n+1} = \mathbf{U}^n + \delta\mathbf{U}$ and $\boldsymbol{\lambda}^{n+1} = \boldsymbol{\lambda}^n + \delta\boldsymbol{\lambda}$):

$$\underbrace{\left[\begin{array}{c|c} \frac{\partial\mathbf{F}_{\text{int}}}{\partial\Delta\mathbf{U}} - \frac{\partial\mathbf{F}_{\text{ext}}^0}{\partial\mathbf{U}} - \eta^n \frac{\partial\mathbf{F}_{\text{ext}}^1}{\partial\mathbf{U}} & \mathbf{B}^T \\ \hline \mathbf{B} & \mathbf{0} \end{array} \right]}_{\mathbf{K}} \underbrace{\begin{bmatrix} \frac{\partial\mathbf{U}}{\partial\boldsymbol{\lambda}} \\ \frac{\partial\boldsymbol{\lambda}}{\partial\boldsymbol{\lambda}} \end{bmatrix}}_{\mathbf{R}^0} = \underbrace{\begin{bmatrix} \mathbf{F}_{\text{ext}}^0 - \mathbf{F}_{\text{int}}^n - \mathbf{B}^T\boldsymbol{\lambda}^n \\ \mathbf{U}_{\text{imp}}^0 - \mathbf{B}\mathbf{U}^n \end{bmatrix}}_{\mathbf{R}^0} + \eta \underbrace{\begin{bmatrix} \mathbf{F}_{\text{ext}}^1 \\ \mathbf{U}_{\text{imp}}^1 \end{bmatrix}}_{\mathbf{R}^1} \quad (2)$$

By solving the linear system (2), the unknowns are reduced to η only:

$$\begin{bmatrix} \delta\mathbf{U} \\ \delta\boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \delta\mathbf{U}^0 \\ \delta\boldsymbol{\lambda}^0 \end{bmatrix} + \eta \begin{bmatrix} \delta\mathbf{U}^1 \\ \delta\boldsymbol{\lambda}^1 \end{bmatrix} \quad \text{where} \quad \begin{bmatrix} \delta\mathbf{U}^0 \\ \delta\boldsymbol{\lambda}^0 \end{bmatrix} = \mathbf{K}^{-1}\mathbf{R}^0 \quad \text{and} \quad \begin{bmatrix} \delta\mathbf{U}^1 \\ \delta\boldsymbol{\lambda}^1 \end{bmatrix} = \mathbf{K}^{-1}\mathbf{R}^1 \quad (3)$$

Finally, η is solution of a single scalar equation, the constraint equation, resulting of the substitution of (3) into the third equation of (1):

$$\tilde{P}(\eta) \underset{\text{definition}}{=} P(\Delta\mathbf{U}^n + \delta\mathbf{U}^0 + \eta\delta\mathbf{U}^1, \eta) \underset{\text{equation}}{=} \Delta\tau \quad (4)$$

It should be noted that the number of solutions to (4), at a given iteration n , is not related to the effective number of solutions to the whole set of equations (1). In particular, even though (1) admits a single solution, it may happen that during the iterative process, i.e. for given $\Delta\mathbf{U}^n$, $\delta\mathbf{U}^0$ and $\delta\mathbf{U}^1$, there is either zero or several solutions η to (4). The latter case is not contradictory since none of these solutions fulfils the equilibrium equations in (1).

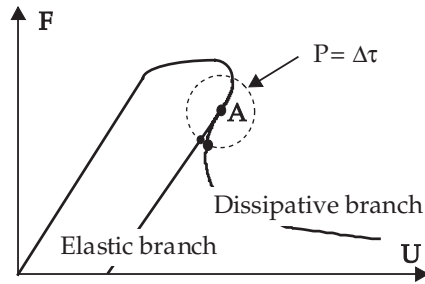


Figure 1. Elastic and dissipative branches are not differentiated with an arc-length method.

2.2. Path-following constraints

The efficiency of the method is highly dependent on the choice of the control function P . The simplest form is a linear function, depending on $\Delta \mathbf{U}$ only:

$$P(\Delta \mathbf{U}) = \mathbf{L} \cdot \Delta \mathbf{U} \quad (5)$$

where $\mathbf{L}$ is a linear form. Equation (4) becomes a linear scalar equation. Unfortunately, when snap-backs occur, it is not always possible to find $\mathbf{L}$ so that (1) admits a solution. Even though strategies have been proposed in Reference [14] to adapt the linear form $\mathbf{L}$ for each increment, the existence of solution is not necessarily guaranteed. This question will be illustrated in Section 6. Therefore, linear constraints do not satisfy our need for *a priori* robustness.

Another choice of path-following constraint is the arc-length one, originally proposed in References [9, 10], where P is a quadratic function of the displacements and of the applied forces:

$$P(\Delta \mathbf{U}, \eta) = \|\Delta \mathbf{U}\|^2 + \underbrace{\beta \|\Delta(\mathbf{F}_{\text{ext}}^0 + \eta \mathbf{F}_{\text{ext}}^1)\|^2}_{\substack{\text{with } \beta \geq 0 \text{ a} \\ \text{scaling parameter}}} \quad (6)$$

For practical reasons, several authors suggest making the scaling parameter β equal to zero, see Reference [12], so that P reduces to a function of the displacement increment only. It has definitely proven its reliability in the context of geometrically non linear problems which exhibit buckling instabilities. However, as initially noticed in Reference [8], it may fail when dealing with material instabilities because a global norm of the displacement is not well-suited to capture localized phenomena resulting from strain localization. Another explanation has been given in Reference [13], the selection of maybe the inadequate solution in (4), and these authors have proposed to select at each iteration (2) the solution that minimizes the residual forces, i.e. the right-hand side in (2). With such a strategy, it seems to be possible to retrieve numerical results that exhibit sharp snap-backs in the context of softening interface laws, see Reference [5].

Yet, on the basis of our own numerical experience of the arc-length method, we relate its inadequacy to the fact that such a path-following constraint does not distinguish the elastic unloading path from a dissipative one, see Figure 1, so that the solution algorithm may capture the elastic solution only, which is not interesting. In consequence, as both paths in Figure 1 realize the equilibrium (thus having the residual forces equal to zero), it is not clear why the

selection proposed in Reference [13] does prevent from selecting the undesirable elastic one. Actually, it does not, as confirmed in Section 4.

2.3. A constraint based on the maximal elastic predictor for the yield function (MEP)

On the basis of the previous observations in the context of material instabilities, it seems that an efficient path-following constraint should:

- preserve the existence of solution,
- not select an elastic unloading path.

An elegant choice would be the energy dissipated by the structure during the current increment. In that way, only dissipating branches of solutions would be selected. However, the dissipation is not a regular function with respect to the displacements and its computation would require the integration of the constitutive law. Therefore, keeping in mind the ingredients of the dissipation function, we propose a simpler control function P .

We depart from the obvious observation that a dissipative path is characterized by the fact that the internal variables evolve somewhere in the structure. If we restrict our attention to constitutive laws governed by a threshold, it means that, somewhere, this threshold is crossed during the elastic prediction phase (positive elastic predictor for the yield function). Then, the location of the dissipation source, the ‘somewhere’, is dealt by considering the maximum of the elastic predictors over the whole structure: it is positive for a dissipative path. This defines the maximal elastic predictor (MEP) path-following constraint.

More precisely, the following general framework for rate-independent materials is considered. The material state is defined by the strain $\boldsymbol{\varepsilon}$ and a set of internal variables a . $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}, a)$ and $A(\boldsymbol{\varepsilon}, a)$ denote the stress and the set of driving forces associated to a . The evolution of a is governed by a yield function $f(A, \boldsymbol{\varepsilon}, a)$ and a flow function $G(A, \boldsymbol{\varepsilon}, a)$ in the following way, where a dot denotes time differentiation:

$$\dot{a} = \lambda G(A, \boldsymbol{\varepsilon}, a) \quad \text{with } \lambda \geq 0 \quad f(A, \boldsymbol{\varepsilon}, a) \leq 0 \quad \lambda f(A, \boldsymbol{\varepsilon}, a) = 0 \quad (7)$$

Then, the elastic predictor, i.e. the value of the yield function with respect to an incrementally elastic response of the material, is defined by

$$f^{\text{el}}(\boldsymbol{\varepsilon}) = f(A(\boldsymbol{\varepsilon}, a^-), \boldsymbol{\varepsilon}, a^-) \quad (8)$$

where the superscript minus denotes quantities at the beginning of the increment. The MEP path-following constraint is defined as the maximum of f^{el} over every Gauss point g , which depends on $\Delta \mathbf{U}$ only:

$$P(\Delta \mathbf{U}) = \max_g f^{\text{el}}(\boldsymbol{\varepsilon}_g^- + \mathbf{B}_g \Delta \mathbf{U}) \quad \text{with } \Delta \boldsymbol{\varepsilon}_g = \mathbf{B}_g \Delta \mathbf{U} \quad (9)$$

where $\mathbf{B}_g$ denotes the derivatives of the shape functions evaluated at Gauss point g .

A possible drawback of (9) lies in its dependence on the constitutive law, implying that a specific development is needed for each constitutive behaviour. A more general function can be proposed, based on the maximal strain increment:

$$P(\Delta \mathbf{U}) = \max_g \frac{\boldsymbol{\varepsilon}_g^-}{\|\boldsymbol{\varepsilon}_g^-\|} \cdot \Delta \boldsymbol{\varepsilon}_g \quad (10)$$

In this case, occurrence of dissipation is approximately characterized by a strain increase somewhere in the structure. However, this characterization is not exact: neither the distinction between an elastic and a dissipative response nor the existence of solution are preserved. Even though this constraint should be efficient in most situations, it may fail for some applications, see Section 6.

Some relations with the framework proposed in Reference [14] should be pointed out. Using the terminology of this paper, the MEP path-following constraint appears as an indirect local sub-plane method (ILSP) where the set of control functions is actually the set of elastic predictors. Nevertheless, there is an important difference in the selection of the active sub-plane: in Reference [14], it seems to be set once and for all at the beginning of the increment as ‘the least critical sub-plane’, whereas it is updated at each iteration through the maximization operator in (9). In that way, at convergence, the active sub-plane does never exhibit limit points, ‘one of the most important problems in path-following techniques’ according to Reference [14].

2.4. Properties of the MEP path-following constraint

First, it is easy to prove that a positive MEP control function selects a dissipative path (since it has been defined for that purpose). A solution is dissipative as soon as at least one gauss point exhibits a positive elastic predictor for the yield function. Therefore, imposing $P(\Delta \mathbf{U}) > 0$ does select a dissipative path:

$$P(\Delta \mathbf{U}) > 0 \Rightarrow \exists g \quad f^{\text{el}}(\mathbf{e}_g) > 0 \Rightarrow \text{dissipative solution} \quad (11)$$

Now, we would like to prove that the MEP path-following constraint does preserve (dissipative) solutions to the non-linear system (1). This is not a trivial question: because of the coupling between the Gauss points through the equilibrium equations, imposing a step-length $\Delta \tau$ too large may preclude the existence of solutions. Actually, the proof is quite technical; we restrict our attention to a necessary condition of existence which ensures that solutions exist for any small enough positive step-length $\Delta \tau$. Mathematically, it reads:

Necessary condition for the existence of solutions. Consider a dissipative branch of solutions $\mathbf{U}(s)$, where $s \in \mathbb{R}^+$ is a parameter strictly increasing with the dissipation (the dissipation itself for instance) and $\mathbf{U}(0) = \mathbf{U}^-$. Then, it is required that:

$$\exists \kappa > 0 \quad \hat{P} : \begin{cases} [0, \kappa] \rightarrow \mathbb{R} \\ s \mapsto P(\mathbf{U}(s) - \mathbf{U}^-) \end{cases} \quad \text{strictly increasing and } \hat{P}(0) = 0 \quad (12)$$

In that case, it is licit to replace s by $\Delta \tau$ as a parameter to describe the dissipative branch from the current state to κ . However, it is not a sufficient condition because κ , or equivalently the size of the maximal step-length $\Delta \tau = \hat{P}(\kappa)$, is not controlled and may go to zero, so that nothing ensures that the whole dissipative branch can be followed by incremental steps, each verifying $P(\Delta \mathbf{U}) = \Delta \tau$. It is only a local condition.

Proposition 1

The necessary condition (12) is fulfilled by the control function P defined in (9) as soon as the constitutive behaviour verifies

$$H = - \left(\frac{\partial f}{\partial A} \cdot \frac{\partial A}{\partial a} + \frac{\partial f}{\partial a} \right) \cdot G > 0 \quad (13)$$

Demonstration

Consider the dissipative branch of solutions $\mathbf{U}(s)$ with $\mathbf{U}(0) = \mathbf{U}^-$. First, let us introduce a compact notation: for any quantity q along the branch, we denote $\hat{q}$ the function that associates q to s . Now, we make a partition of the Gauss points in two sets: one set E of elastic Gauss points and one set D of dissipative Gauss points, depending on the plastic multiplier λ_g :

$$\begin{aligned} E &= \{g; \exists s_g > 0 \quad \forall s \in]0, s_g[\quad \hat{\lambda}_g(s) = 0\} & \text{and} \quad \kappa = \min_g s_g > 0 \\ D &= \{g; \exists s_g > 0 \quad \forall s \in]0, s_g[\quad \hat{\lambda}_g(s) > 0\} \end{aligned} \quad (14)$$

where the fact that the number of Gauss points is finite is essential. As the branch is dissipative, D is not empty. For elastic Gauss points $g \in E$, we have

$$\forall s \in]0, \kappa[\quad \hat{\lambda}_g(s) = 0 \Rightarrow \hat{a}_g(s) = a_g^- \Rightarrow \hat{f}_g^{\text{el}}(s) = \hat{f}_g(s) \leq 0 \quad (15)$$

For dissipative Gauss points $g \in D$, taking into account the flow rule, we have

$$\frac{d\hat{f}}{ds} = \left(\frac{\partial f}{\partial A} \cdot \frac{\partial A}{\partial \varepsilon} + \frac{\partial f}{\partial \varepsilon} \right) \cdot \frac{d\hat{\varepsilon}}{ds} + \left(\frac{\partial f}{\partial A} \cdot \frac{\partial A}{\partial a} + \frac{\partial f}{\partial a} \right) \cdot \frac{d\hat{a}}{ds} = \frac{d\hat{f}^{\text{el}}}{ds} - \hat{\lambda} \hat{H}$$

Thanks to definition (14) and the consistency condition, we deduce

$$\forall s \in]0, \kappa[\quad \frac{d\hat{f}}{ds}(s) = 0 \Rightarrow \frac{d\hat{f}_g^{\text{el}}}{ds}(s) = \hat{\lambda}_g(s) \hat{H}_g(s) \quad \text{with} \quad \hat{\lambda}_g(s) > 0$$

Therefore, the strict positivity of H , i.e. condition (13), implies that f_g^{el} is strictly increasing in $]0, \kappa[$ and

$$\forall s \in]0, \kappa[\quad \hat{f}_g^{\text{el}}(s) > \hat{f}_g^{\text{el}}(0) = \hat{f}_g(0) = 0 \quad (16)$$

Now, one can come back to P . Thanks to (15) and (16), one has

$$\hat{P}(0) = \max_g \hat{f}_g(0) = 0 \quad \text{and} \quad \hat{P}(s) = \max_g \hat{f}_g^{\text{el}}(s) = \max_{g \in D} \hat{f}_g^{\text{el}}(s) \quad \text{strictly increasing} \quad \square$$

Independently of path-following, condition (13) is an essential requirement for the constitutive law since it ensures that the response to any given strain rate exists and is a unique stress rate and internal variable rate. Already proposed in Reference [17] for plasticity and in Reference [18] for brittle damage, it is a necessary and sufficient condition for the plastic multiplier λ obtained through the consistency condition to be positive. Indeed, it is given by the following expression (where $\langle \cdot \rangle$ are McAuley's brackets and denote the positive part function):

$$\lambda = \frac{1}{H} \left\langle \left(\frac{\partial f}{\partial A} \cdot \frac{\partial A}{\partial \varepsilon} + \frac{\partial f}{\partial \varepsilon} \right) \cdot \dot{\varepsilon} \right\rangle \quad (17)$$

3. NUMERICAL SOLUTION OF THE MEP PATH-FOLLOWING EQUATION

To determine the load factor η by means of the method recalled in Section 2.1, the constraint equation (4) has to be solved. In the case of the MEP control function (9), $\tilde{P}(\eta)$ is defined as

the maximum of the contributions $\tilde{f}_g^{\text{el}}(\eta)$ of each Gauss points g :

$$\tilde{P}(\eta) = \max_g \tilde{f}_g^{\text{el}}(\eta) \quad \text{with} \quad \tilde{f}_g^{\text{el}}(\eta) = f_g^{\text{el}}(\boldsymbol{\epsilon}_g^- + \mathbf{B}_g(\Delta \mathbf{U}^n + \delta \mathbf{U}^0 + \eta \delta \mathbf{U}^1)) \quad (18)$$

There are two sources of non-linearity in the function $\tilde{P}$: the first one comes from the elastic predictors f^{el} themselves, the second one results from the function max which is even not differentiable. Because of this latter characteristic, straightforward applications of usual numerical methods, Newton's one for instance, are doomed to failure. Therefore, we propose a specific method which splits the difficulties:

- each function $\tilde{f}_g^{\text{el}}$ is linearized in some way and a piecewise linear approximation Π of $\tilde{P}$ is defined as the maximum of these linear functions;
- the equation $\Pi(\eta) = \Delta\tau$ is solved instead of the true path-following equation $\tilde{P}(\eta) = \Delta\tau$.

Finally, thanks to an appropriate choice of linearization along with a characteristic property of the function max, the approximated equation and the true one admit the same solutions. Actually, this is true only if f^{el} is a convex function of the strain tensor. As it encompasses most of the usual constitutive relations, this property is assumed from now on, which implies that $\tilde{f}_g^{\text{el}}$ is convex with respect to η , thanks to (18) (conjugacy of a convex map with a linear one).

3.1. Definition of the approximated control function

To gain the remarkable property that both sets of solutions coincide, each function $\tilde{f}_g^{\text{el}}$ has to be linearized in the neighbourhood of the points η characterized by the following local equation:

$$\tilde{f}_g^{\text{el}}(\eta) = \Delta\tau \quad (19)$$

This equation depends on the constitutive relation and therefore requires a specific treatment. We assume that it can be solved efficiently, with closed-form solutions or Newton's method for instance. Generally, it admits zero, one or two solutions.

The case with zero solution can be divided in two sub-cases whether $\tilde{f}_g^{\text{el}}$ (which is continuous since it is convex, proper and depends on a real variable) stays below or above $\Delta\tau$.

- $\Delta\tau$ is an upper bound of $\tilde{f}_g^{\text{el}}$.
As $\tilde{f}_g^{\text{el}}$ is convex, it implies that it is a constant function. The considered Gauss point is useless to the path-following method and should not be taken into account. If every Gauss point is useless, then the load does not trigger a dissipative solution, thanks to Proposition 1.
- $\Delta\tau$ is a lower bound of $\tilde{f}_g^{\text{el}}$.

The path-following equation (4) does not admit any solution since $\tilde{P}(\eta) > \Delta\tau$. This case may be related to the fact that the increment $\Delta\tau$ is too large, in the same way as Reference [19] did when considering the lack of real roots for the arc-length path-following equation. Therefore, we suggest to subdivide the increment in smaller ones and to restart the solution of (1) from the beginning, as will be explained in more details hereafter.

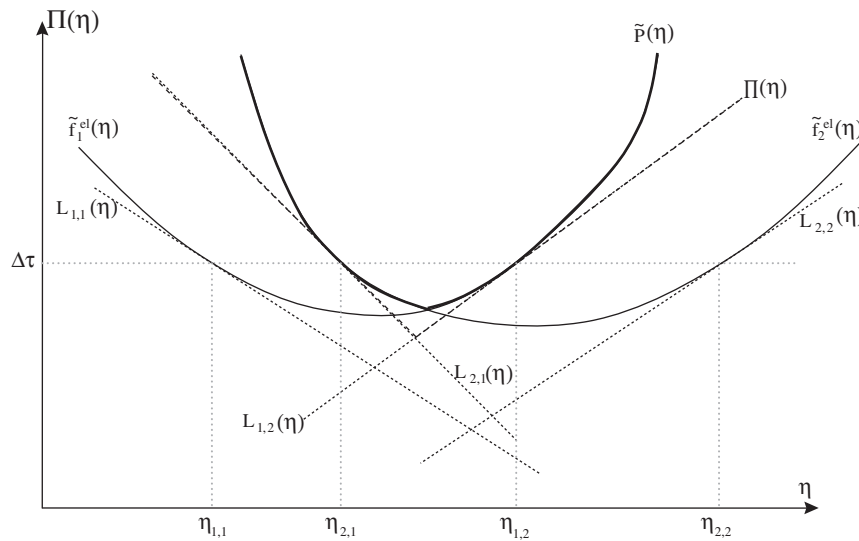


Figure 2. Approximation function Π .

We can now consider that (19) admits solutions, say $\eta_{g,\alpha}$, with $\alpha \in \{1, 2\}$ depending on the effective number of solutions. Then, the following linearized parts of $\tilde{f}_g^{\text{el}}$ are introduced:

$$L_{g,\alpha}(\eta) = \tilde{f}_g^{\text{el}}(\eta_{g,\alpha}) + \frac{\text{d}\tilde{f}_g^{\text{el}}}{\text{d}\eta}(\eta_{g,\alpha})(\eta - \eta_{g,\alpha}) = A_{g,\alpha}^{(0)} + A_{g,\alpha}^{(1)}\eta \quad \text{with } A_{g,\alpha}^{(1)} \neq 0 \quad (20)$$

Finally, we define the approximated control function Π , see Figure 2:

$$\Pi(\eta) = \max_{g, \alpha} L_{g, \alpha}(\eta) \quad (21)$$

3.2. Properties of the approximated control function

A brief study of the properties of the approximated control function Π shows that it is less than the true control function $\tilde{P}$ and that $\Pi(\eta) = \Delta\tau$ and $\tilde{P}(\eta) = \Delta\tau$ admit the same set of solutions. The assumed convexity of f^{el} is essential here.

Proposition 2

Π is less than $\tilde{P}$:

$$\forall \eta \quad \Pi(\eta) \leq \tilde{P}(\eta) \quad (22)$$

Demonstration

Thanks to the convexity of $\tilde{f}_g^{\text{cl}}$, the following inequality is achieved for the local contributions, hence (22):

$$L_{g,\alpha}(\eta) \leq \tilde{f}_g^{\text{el}}(\eta)$$

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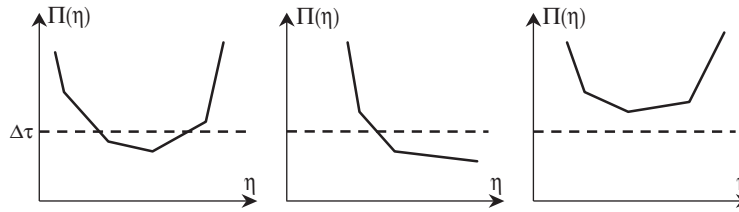


Figure 3. The various cases for Equation (24): two, one or no solutions.

Proposition 3

The sets of solutions for $\tilde{P}(\eta) = \Delta\tau$ and $\Pi(\eta) = \Delta\tau$ are equal:

$$\tilde{P}(\eta) = \Delta\tau \Leftrightarrow \Pi(\eta) = \Delta\tau \quad (23)$$

Demonstration

First, suppose that $\tilde{P}(\eta) = \Delta\tau$. Then:

$$\exists g \quad \tilde{P}(\eta) = \tilde{f}_g^{\text{el}}(\eta) = \Delta\tau \Rightarrow \exists g, \alpha \quad \eta = \eta_{g,\alpha} \quad \text{and} \quad L_{g,\alpha}(\eta) = \Delta\tau$$

Moreover, thanks to (22), $\Pi(\eta) \leq \tilde{P}(\eta)$. Therefore

$$\exists g, \alpha \quad \Delta\tau = L_{g,\alpha}(\eta) \leq \Pi(\eta) \leq \tilde{P}(\eta) = \Delta\tau \Rightarrow \Pi(\eta) = \Delta\tau$$

Conversely, suppose now that $\Pi(\eta) = \Delta\tau$. Thanks again to the convexity of $\tilde{f}_g^{\text{el}}$, one has for a given η :

$$[\forall \alpha \quad L_{g,\alpha}(\eta) \leq \Delta\tau] \Rightarrow \tilde{f}_g^{\text{el}}(\eta) \leq \Delta\tau$$

Taking advantage of this implication, one obtains

$$\Pi(\eta) = \Delta\tau \Rightarrow \forall g, \alpha \quad L_{g,\alpha}(\eta) \leq \Delta\tau \Rightarrow \forall g \quad \tilde{f}_g^{\text{el}}(\eta) \leq \Delta\tau \Rightarrow \tilde{P}(\eta) \leq \Delta\tau$$

As $\Pi(\eta) \leq \tilde{P}(\eta)$, it implies that $\tilde{P}(\eta) = \Delta\tau$.

3.3. Solution of the approximated constraint equation

Now, we can take advantage of Proposition 3: the equation

$$\Pi(\eta) = \Delta\tau \quad (24)$$

is solved instead of $\tilde{P}(\eta) = \Delta\tau$. As already noticed in Section 2.1, the number of solutions to (4), or equivalently to (24), is not related to the effective number of solutions to the whole system (1). Actually, Equation (24) admits zero, one or two solutions, see Figure 3, because Π is (strictly) convex. Therefore, a strategy has to be adopted whenever the cases with zero or two solutions are encountered.

- When (24) admits two solutions, one can select the one which makes the residual forces in (2) minimal, as suggested in Reference [13]. Such a choice proves to be stabilizing regarding the convergence of the Newton–Raphson algorithm and therefore worth its numerical cost.

Box I. Algorithm to solve the scalar load control equation.

(1)	Initialization of the interval	$I :=] - \infty + \infty[$
(2)	Loop over the linear functions g, α	
(2.1)	Solution of the active linear function	$\eta_{g,\alpha}$ s.t. $L_{g,\alpha}(\eta_{g,\alpha}) = \Delta\tau$
(2.2)	Setting of the new interval	
(2.2.1)	if the active linear function is increasing	$A_{g,\alpha}^{(1)} > 0 \Rightarrow I := I \cap] - \infty, \eta_{g,\alpha}]$
(2.2.2)	if the active linear function is decreasing	$A_{g,\alpha}^{(1)} < 0 \Rightarrow I := I \cap [\eta_{g,\alpha}, \infty[$
(2.3)	Exit if the interval is empty	
(3)	The solutions are the bounds of the interval	$\eta \in \text{Fr}(I) \Rightarrow \Pi(\eta) = \Delta\tau$

- When there is no solution, we suggest once more to subdivide the increment, as for the local equation (19), because we believe that the lack of solutions may be related to too large increments, as in Reference [19].

A possible algorithm to solve Equation (24) when at least one solution exists is presented in Box I. It is based on the construction of successive nested intervals: the bounds of the last one are the solutions of the equation. It is very fast since it relies on G solutions of scalar linear equations, where G is the total number of Gauss points. It may stop before reaching the end if the intervals become empty, that is when Equation (24) does not admit a solution.

In conclusion, the solution of the MEP constraint equation relies on local solutions of (usually) simple equations which explicitly depend on the constitutive law, followed by a global solution based on the general algorithm of Box 1. Several applications are now going to illustrate its potency along with the interest of the MEP path-following constraint.

4. APPLICATION TO A COHESIVE CRACK MODEL

The first application concerns the numerical simulation of crack propagations through a perforated plate which is submitted to a tension applied on both ends, see Figure 4. The crack propagation is governed by Barenblatt's law [20], which models the process-zone ahead from the crack tip. A rigorous analysis is given in Reference [21] in terms of a variational formulation where the position of the cracks is considered as an unknown. However, we restrict our attention to a simpler case by postulating *a priori* the potential crack paths along the symmetry line A–A' right and left from the hole. As the crack paths are set from the beginning, classical interface finite elements are used to model the cracks, so that Barenblatt's law appears as an interface law which can be cast in the framework (7). The MEP path-following method is applied to overcome the numerous snap-backs that are expected with such a brittle interface law, see for instance Reference [5].

4.1. Barenblatt's law

In mode I propagation, Barenblatt's fracture law relates the normal displacement discontinuity along the crack lips $[u_n]$ to the applied normal tension σ_n :

$$\sigma_n = \sigma_c \exp\left(-\frac{\sigma_c}{G_f} [u_n]\right) \quad (25)$$

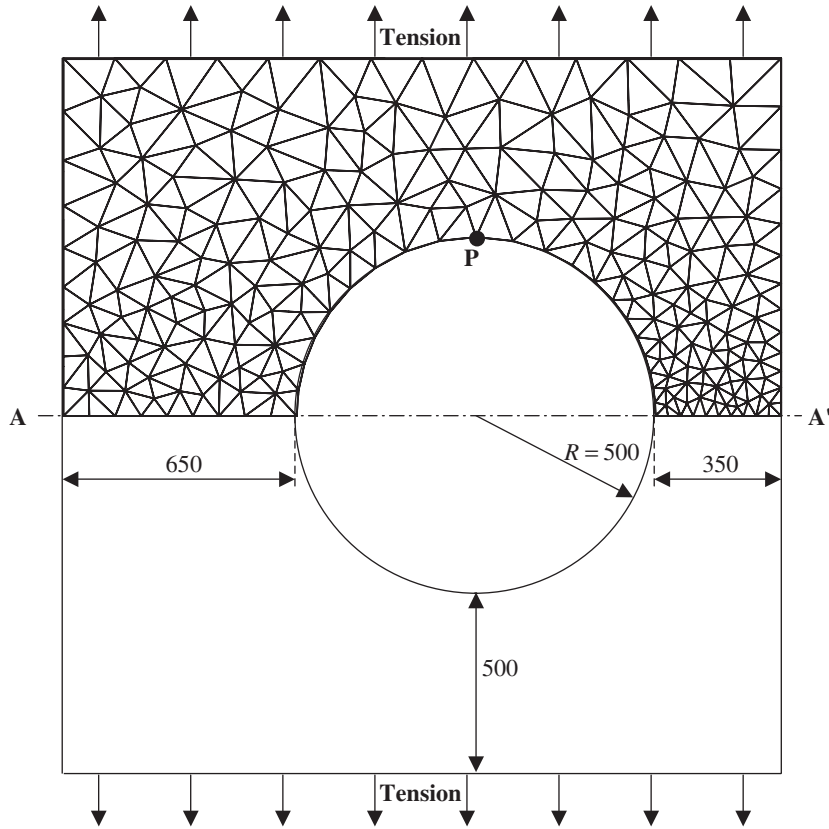


Figure 4. Geometry of the perforated plate and load conditions (dimensions in mm).

where G_f is the fracture surface energy and σ_c the critical stress. Applying this law in structural computations requires in addition to treat potential unloading. For that purpose, Reference [22] introduced a new internal variable a , whose evolution is governed by a threshold f , a consistency condition and a flow rule:

$$f([u_n], a) = [u_n] - a \quad \text{with} \quad \begin{cases} f \leq 0 & \lambda \geq 0, \quad \lambda f = 0 \\ \dot{a} = \lambda \end{cases} \quad (26)$$

Moreover, relation (25) now becomes

$$\sigma_n = \sigma_c \exp\left(-\frac{\sigma_c}{G_f} a\right) \frac{[u_n]}{a} \quad (27)$$

Finally, the contact condition between the crack lips simply reads

$$[u_n] \geq 0 \quad (28)$$

Thus, the model exhibits linear unloading and dissipative softening according to the stress–displacement discontinuity relation (25), see Figure 5.

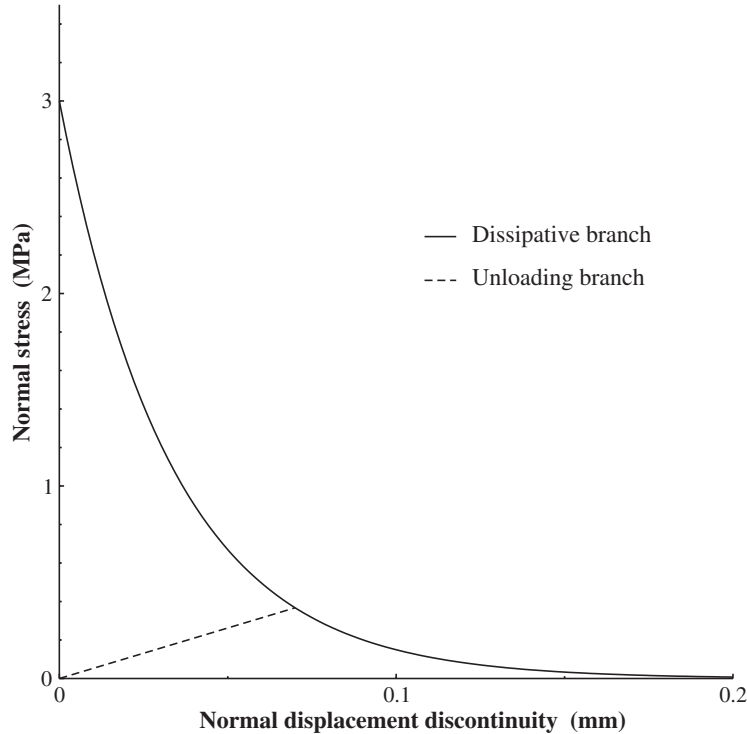


Figure 5. Barenblatt's law.

The evolution equation (26) belongs to framework (7), where the displacement discontinuity plays the role of the strain. Moreover, thanks to the use of interface finite elements along the crack paths, the displacement discontinuity is expressed as a linear function of the nodal displacements at each integration point g of these elements:

$$[u_n]_g = \mathbf{B}_g^{\text{CZM}} \mathbf{U} \quad (29)$$

Thus, the MEP path-following method is fully available to perform crack propagation computations. A normalized elastic predictor for the yield function simply reads

$$f^{\text{el}}([u]) = \frac{[u]}{a^-} - 1 \quad (30)$$

4.2. Numerical results

The material parameters, elasticity constants for the plate and Barenblatt's law coefficients, are given in Table I. The numerical simulation is performed under a plane strain assumption with a 2D unstructured mesh obtained with GMSH software [23]. It takes advantage of the symmetry and it is composed of linear finite elements for the elastic zone (412 triangles with 3 nodes and one Gauss point) and linear interface finite elements for the crack paths (19

Table I. Material parameters (simulation with Barenblatt's law).

Elasticity	Barenblatt's law
$E = 20\,000 \text{ MPa}$	$\sigma_c = 3 \text{ MPa}$
$\nu = 0.2$	$G_f = 0.1 \text{ N mm}^{-1}$

degenerated 4-node quadrangles). Besides, the contact condition (28) is handled with a penalty method while the non-definiteness of the model when $a = 0$ is avoided by regularizing the infinite initial stiffness of the cohesive zone through the introduction of a small initial threshold $a(0) = a^0$. Even though the accuracy of the results is expected to be rather poor (linear shape functions, penalty coefficient, stiffness regularization), this application nevertheless highlights the robustness of the MEP path-following constraint.

The computation is achieved with the finite element software *Code_Aster* [24]. The temporal integration of the constitutive law is based on Euler's backward scheme. A convergence criterion of 10^{-2} N on the residual forces (1) is adopted. It appears as a reasonable compromise between time cost and accuracy:

$$\|\mathbf{F}_{\text{int}}(\Delta\mathbf{U}) + \mathbf{B}^T \lambda - \mathbf{F}_{\text{ext}}^0(\mathbf{U}) - \eta \mathbf{F}_{\text{ext}}^1(\mathbf{U})\|_{L^\infty} \leq 10^{-2} \quad (31)$$

Finally, the initial step-length is set to $\Delta\tau = 0.1$. Each time the convergence is not achieved within seven iterations of the Newton–Raphson algorithm or each time the path-following equation does not admit any solution, the current increment is divided in four sub-increments. This strategy appears quite crude and more refined ones may be applied. Nevertheless, it is robust thanks to the continuity property (12) which ensures that when the increment size $\Delta\tau$ goes to zero, the unknown $(\mathbf{U}, \lambda, \eta)$ tend to their values at the beginning of the increment, thus making the solution easier for Newton's method. The computation is stopped as soon as the applied tension goes below 0.01 MPa. The whole computation consists of 784 increments; only few of them require subdivisions (total number of increments and sub-increments equal to 841, no more than seven sub-increments for a single increment). This desirable behaviour is related to the selection of the solution proposed in Reference [13] since it avoids numerical oscillations of the algorithm between the solutions to the constraint equation. Indeed, the same computation was performed with another selection criterion (η which leads to the smallest angle between the previous and the current incremental displacement): it required $\Delta\tau = 0.001$ to get convergence, so that the total number of increments would have reached about 100 000, had the computation not been halted before the end.

The global response of the structure is given in Figure 6 in terms of the applied tension versus the opening displacement (vertical enlargement of the hole, equal to twice the vertical displacement of the point **P** in Figure 4). Six remarkable points (**O**, **P**, **A**, **B**, **C** and **E**) are defined on the curve. The corresponding normalized crack opening distributions along the crack paths are plotted in Figure 7, where the normalized crack opening $\tilde{a}$ is defined to range from 0 (sound) to 1 (totally cracked):

$$\tilde{a} = 1 - \exp\left(-\frac{\sigma_c}{G_f} a\right) \quad (32)$$

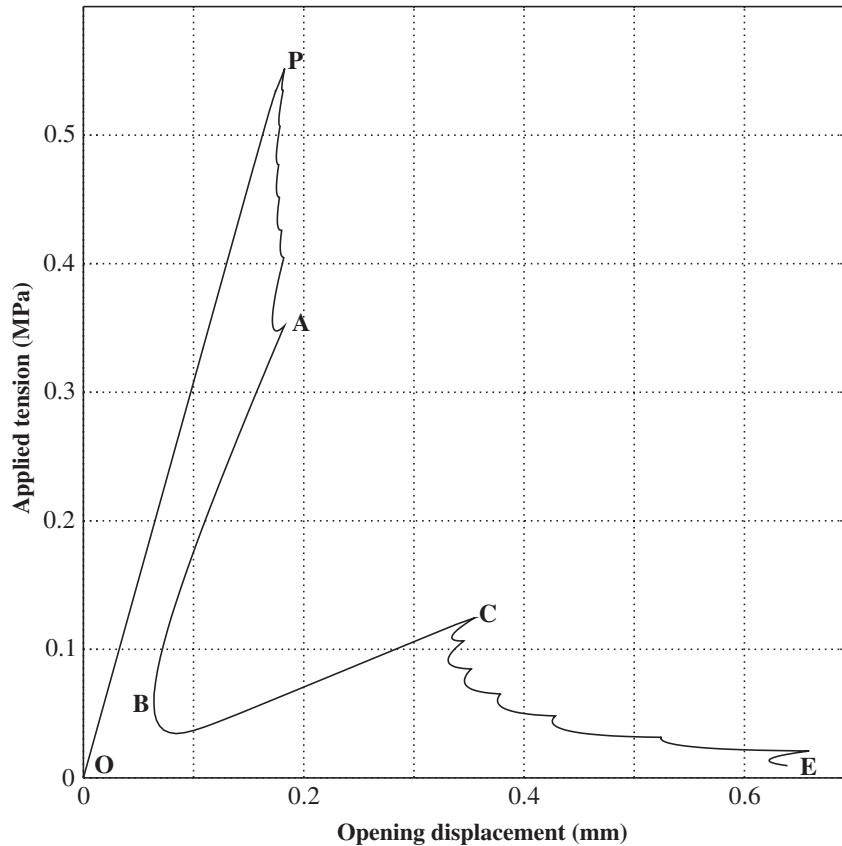


Figure 6. Global response of the perforated plate (CZM formulation).

During the phase $O \rightarrow P$, two cracks propagate left and right from the hole. Then, the left crack stops, the right one goes on propagating ($P \rightarrow A$) and the opening displacement remains nearly constant. When the right free boundary is reached, the crack opening at the right end evolves, resulting in a snap-back ($A \rightarrow B$). Then, the left crack starts opening again ($B \rightarrow C$) before finally propagating ($C \rightarrow E$).

The stability of the solution is deduced from an analysis of the energetic response of the structure, see Figure 8. It is unstable when the power of external forces is negative, that is when some energy is released by the structure. It appears that the phases $O \rightarrow P$ and $B \rightarrow C$ are stable, the phases $P \rightarrow A$ and $C \rightarrow E$ are first unstable then stable, the phase $A \rightarrow B$ is unstable. It can be observed that the instabilities are related to a release of elastic energy larger than the dissipation, inducing a decrease of the work of external forces. Besides, small oscillations of the work of external forces are also observed: they are related to the finite element discretization of the crack paths and correspond to the ongoing damage of one element after the other. These observations are in agreement with other studies, see for instance References [3–5].

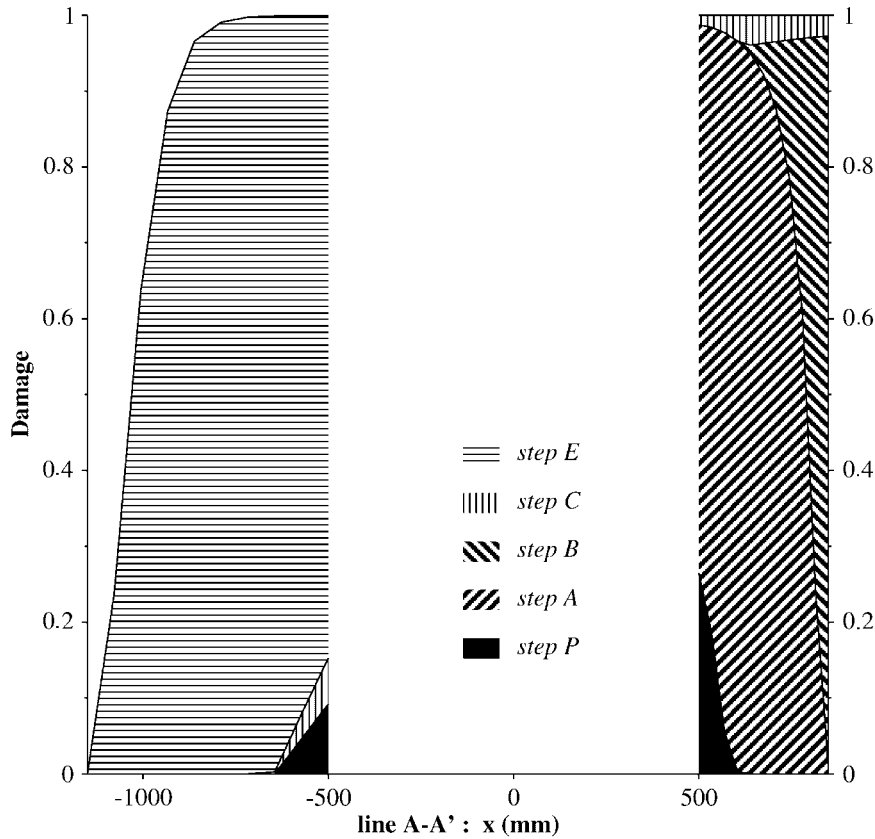


Figure 7. Normalized crack opening along the crack paths.

Regarding the various stages of the response, this simulation definitely proves the robustness of the MEP path-following constraint, at least in the context of softening cohesive zone models. Beyond its ability to follow small as well as large snap-backs, two major facts should be pointed out:

- the phase $A \rightarrow B$ proves that sharp snap-backs do not raise special difficulties, thanks to the selection of a dissipative solution so that its closeness to a pure elastic unloading solution does not mind (actually, another computation was performed with an arc-length method but the elastic solution was always selected as soon as the point P was reached);
- the method allows to follow several cracks and to skip from one to another whenever some of them stop propagating, thanks to the maximization of the elastic predictor which results in seeking a place where dissipation occurs without *a priori* hints.

We think that this robustness is worth the additional cost related to the non linear character of the path-following constraint. Indeed, in the present case, the time spent within the subroutines that manage the solution of this constraint represents only 5% of the total computation time.

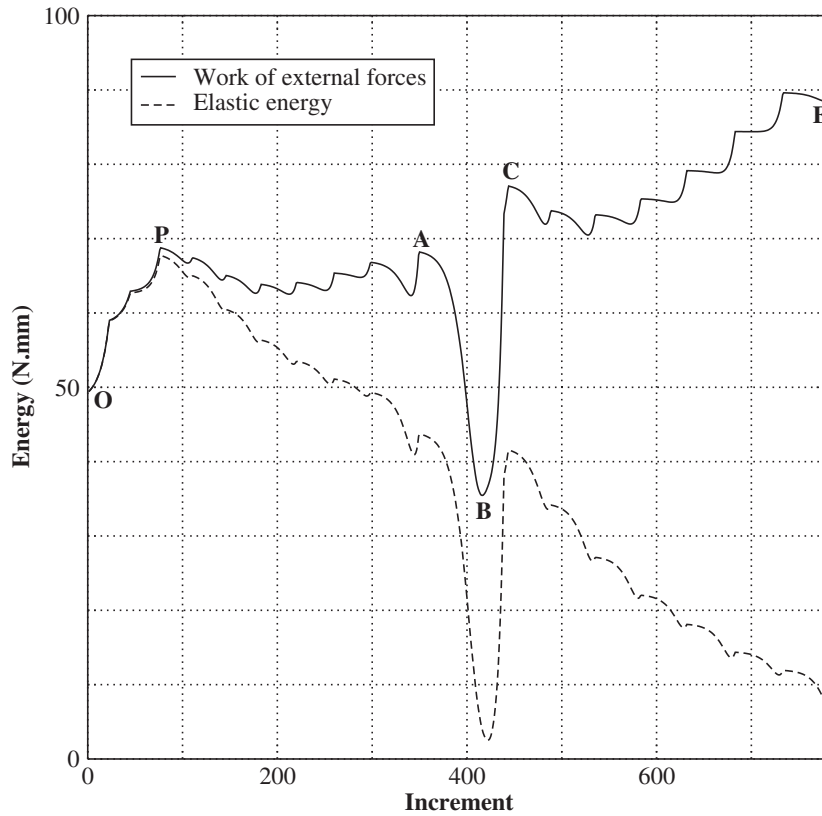


Figure 8. Energy transfers during loading.

5. APPLICATION TO A NON-LOCAL DAMAGE MODEL

In this second application, we focus again on the perforated plate depicted in Figure 4. However, instead of looking for the propagation of cracks along predefined paths, we consider now that the whole material is brittle elastic, its behaviour described by a full 3D damage model. Because of the strain-softening character of the constitutive law, the equations of the problem require some regularization to remain well-posed. Even though this is not the scope of the present article, a non-local formulation is therefore adopted. This quite different problem compared to the crack propagation of Section 4 is thought to illustrate the wide range of applications covered by the MEP path-following method.

5.1. Damage constitutive law

The material state is described through the strain tensor $\boldsymbol{\varepsilon}$ and a scalar damage variable d which increases from 0 (sound material) to 1 (totally damaged material with zero stiffness). The stress $\boldsymbol{\sigma}$ and the driving force A are given by

$$\boldsymbol{\sigma} = (1 - d)\mathbf{E} \cdot \boldsymbol{\varepsilon} \quad \text{and} \quad A = \frac{1}{2} \boldsymbol{\varepsilon} \cdot \mathbf{E} \cdot \boldsymbol{\varepsilon} \quad (33)$$

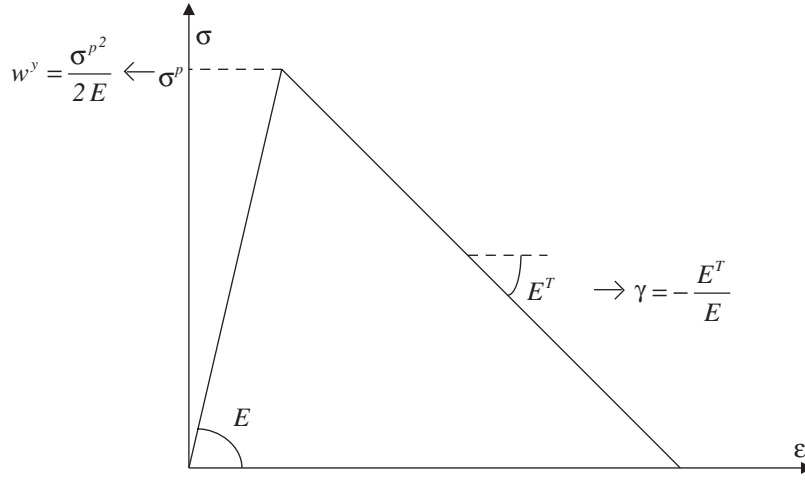


Figure 9. Brittle damage law: simple tension test.

where $\mathbf{E}$ denotes Hooke's isotropic tensor (Young modulus E and Poisson ratio ν). The damage evolution is governed by the following yield function, flow rule and consistency condition:

$$f(A, d) = A - w^y \left(\frac{1 + \gamma}{1 + \gamma - d} \right)^2 \quad (34)$$

$$\dot{d} = \lambda \quad \text{and} \quad \lambda \geq 0 \quad f(A, d) \leq 0 \quad \lambda f(A, d) = 0 \quad (35)$$

w^y characterizes the initial threshold and γ the softening behaviour, see Figure 9 for a simple tension test. Such a model belongs to the general framework (7) with $G = 1$. It fulfils the essential condition (13), ensuring that any strain path is licit:

$$H = - \left(\frac{\partial f}{\partial A} \cdot \frac{\partial A}{\partial a} + \frac{\partial f}{\partial a} \right) \cdot G = \frac{2w^y(1 + \gamma)^2}{(1 + \gamma - d)^3} > 0 \quad (36)$$

The elastic predictor for the yield function is derived by setting the internal variable d to its value at the beginning of the increment d^- . According to (33) and (34), a normalized elastic predictor for the yield function is then

$$f^{\text{el}}(\boldsymbol{\varepsilon}) = \frac{\boldsymbol{\varepsilon} \cdot \mathbf{E} \cdot \boldsymbol{\varepsilon}}{2w^y} - \left(\frac{1 + \gamma}{1 + \gamma - d} \right)^2 \quad (37)$$

f^{el} is convex with respect to $\boldsymbol{\varepsilon}$ (thanks to the positiveness of $\mathbf{E}$), so that the scheme proposed in Section 3 is available: the local equation (19) simply results in solving a quadratic polynomial which does not require any additional numerical scheme.

5.2. Nonlocal formulation

To avoid spurious localization due to strain-softening, see Reference [25] for a review, a non-local formulation based on a smoothed strain is adopted. Inspired by Reference [26],

its description and its finite element implementation are given in Reference [27], while its mathematical properties and some links with other formulations are presented in Reference [28].

In few words, consider a constitutive law which is encompassed within the formalism (7). A smoothed strain tensor $\bar{\epsilon}$ is derived from the usual strain tensor ϵ by means of the following differential regularization operator:

$$\begin{aligned} \bar{\epsilon} - l_c^2 \Delta \bar{\epsilon} &= \epsilon \quad \text{in } \Omega \\ \nabla \bar{\epsilon} \cdot \mathbf{n} &= 0 \quad \text{on } \partial\Omega \end{aligned} \quad (38)$$

where Ω denotes the body domain, $\partial\Omega$ its boundary and $\mathbf{n}$ the corresponding outer normal. l_c is a characteristic length of the material. The local stress-strain relation $\sigma(\epsilon, a)$ is preserved while the smoothed strain tensor is introduced into the definition of the set of driving forces $A(\bar{\epsilon}, a)$ associated to the internal variables and into the evolution equations:

$$\dot{a} = \lambda G(A, \bar{\epsilon}, a) \quad \text{with } \lambda \geq 0 \quad f(A, \bar{\epsilon}, a) \leq 0 \quad \lambda f(A, \bar{\epsilon}, a) = 0 \quad (39)$$

On a numerical ground, specific finite elements are used to solve at the same time the equilibrium equations and the regularization equation (38). Their degrees of freedom are the nodal displacement $\mathbf{U}$ and the nodal smoothed strain tensor $\bar{\mathbf{E}}$. The (discretized) displacement field $\mathbf{u}$ and smoothed strain field $\bar{\epsilon}$ are expressed as

$$\begin{bmatrix} \mathbf{u}(\mathbf{x}) \\ \bar{\epsilon}(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \mathbf{N}^u(\mathbf{x}) & \mathbf{0} \\ \mathbf{0} & \mathbf{N}^\epsilon(\mathbf{x}) \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \bar{\mathbf{E}} \end{bmatrix} \quad (40)$$

where $\mathbf{N}^u$ and $\mathbf{N}^\epsilon$ denote, respectively, the matrices of the quadratic shape functions for the displacement discretization and linear shape functions for the smoothed strain one. Besides, in the context of a brittle damage law as described in the previous sub-section, the contributions of Gauss points where damage has reached its ultimate value $d = 1$ are removed from equation (38) to avoid the spurious enlargement of the damaged zone observed in Reference [29].

Because of (39), the elastic predictor for the yield function is now a function of the smoothed strain tensor instead of the local one. According to (37), it reads

$$f^{\text{el}}(\bar{\epsilon}) = \frac{\bar{\epsilon} \cdot \mathbf{E} \cdot \bar{\epsilon}}{2w^y} - \left(\frac{1 + \gamma}{1 + \gamma - d} \right)^2 \quad (41)$$

Thanks to (40), the smoothed strain tensor is expressed as a linear function of the nodal unknowns. Therefore, the MEP path-following constraint can be defined in the same way as in (9), where $\mathbf{N}^\epsilon$ plays the role of the matrix $\mathbf{B}$:

$$P \left(\begin{bmatrix} \Delta \mathbf{U} \\ \Delta \bar{\mathbf{E}} \end{bmatrix} \right) = \max_g f^{\text{el}}(\bar{\epsilon}_g^- + \mathbf{N}_g^\epsilon \Delta \bar{\mathbf{E}}) \quad (42)$$

In conclusion, it appears that the use of such a non-local formulation is fully compatible with the MEP path-following method to perform numerical simulations.

Table II. Material parameters (simulation with a non-local damage model).

Elasticity	Damage	Nonlocality
$E = 20\,000\text{ MPa}$ $\nu = 0.2$	$\sigma^p = 3\text{ MPa}$ $E^T = -5\,000\text{ MPa}$ $\downarrow$ $w^y = 2.25 \cdot 10^{-4}\text{ MPa}$ $\gamma = 0.25$	$l_c = 15\text{ mm}$

5.3. Numerical results

The material parameters used in the simulation are given in Table II. Once more, the numerical simulation is performed under a plane strain assumption with a 2D unstructured mesh. Compared to Section 4, the mesh has to be refined in zones where damage is expected to gain a reasonable accuracy for the nonlocal treatment (the size of the smaller finite elements is equal to 15 mm). In addition, to avoid selecting *a priori* a symmetrical solution, the whole structure is meshed without taking advantage of the symmetry of the geometry and the loading. Consequently, the mesh is now composed of 2148 quadratic triangles (6 nodes for the displacements, 3 nodes for the smoothed strain and 3 Gauss points).

The computation is achieved with a convergence criterion of 10^{-3}N on the residual forces (smaller than previously because of the use of smaller finite elements in the critical zones). The initial step-length is set to $\Delta\tau = 1$ and the same subdivision strategy as previously is adopted. As the constitutive law is not fit to properly distinguish between traction and compression damage, the computation is stopped as soon as damage in compression occurs (on the left border of the plate), that is after 700 increments. Some of them require several subdivisions, up to 32 sub-increments for a single increment (total number of increments and sub-increments equal to 1318, thus an average of nearly two sub-increments per increment). It should also be noted that the solution of the path-following equation requires only 8% of the total CPU time.

The global response of the structure in terms of applied stress versus opening displacement (vertical enlargement of the hole) is plotted in Figure 10. It exhibits the same qualitative features as with the cohesive crack formulation. Namely, numerous spurious snap-backs are observed, in spite of the regularized character of the non-local law. They may be related to the brutal change of behaviour when a point reaches the ultimate damage $d = 1$. Therefore, they should reduce with a finer mesh: this point will be investigated in a future work about this nonlocal formulation.

The damage field is plotted for three remarkable points **P**, **B** and **E**, mentioned on the global response curve. They correspond, respectively, to:

- the peak load (inception of damage on both sides of the hole), Figure 11;
- the stage when the propagating damaged zone departing from the right side of the hole reaches the right border of the plate, see Figure 12 where it can be checked that the damage band width spreads over several finite elements, thanks to the non-local formulation;
- the final stage when damage in compression occurs, a new damaged zone having propagated from the left of the hole almost to the left border of the plate, Figure 13.

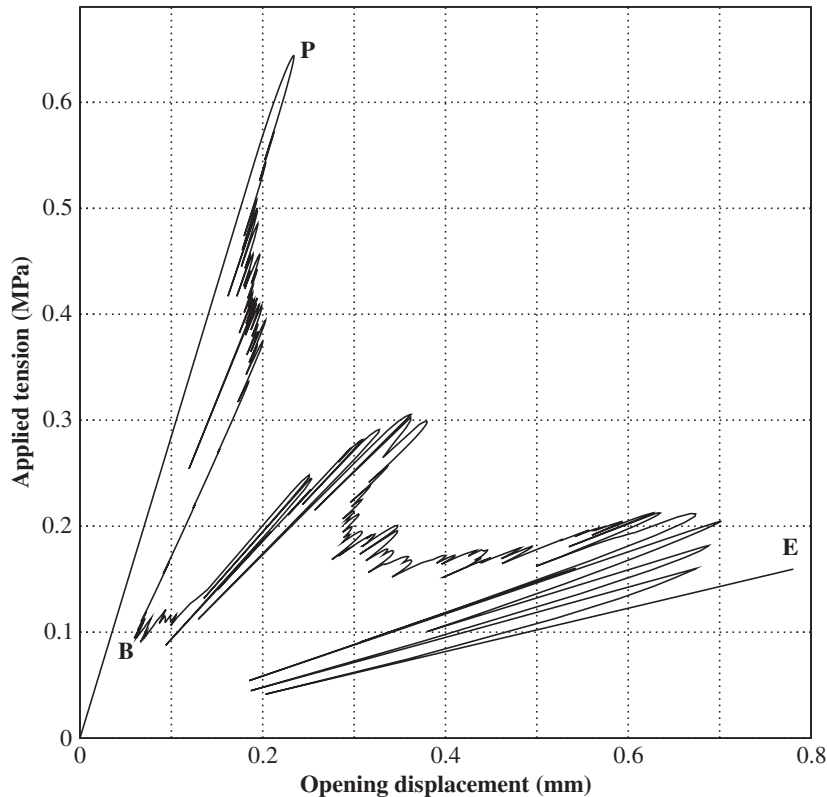


Figure 10. Global response of the perforated plate (non-local formulation).

Finally, this application confirms the good behaviour of the MEP path-following constraint already observed in the previous section, namely its ability to track several damaged zones and to skip automatically from one to another when required, thanks to the max function in (9). Moreover, it seems that the non-local character of the constitutive law does not raise any specific difficulty.

6. APPLICATION TO STRAIN-SOFTENING PLASTICITY

Two applications have already showed the efficiency of the MEP path-following constraint in the context of real-life numerical simulations. A third one based on a homogeneous stress state is presented for several different reasons:

- to demonstrate the degree of generality of the formulation by applying it to strain-softening plasticity;
- to illustrate the existence of snap-backs that are related to the constitutive behaviour itself and not to the structural effect of an elastic energy release;

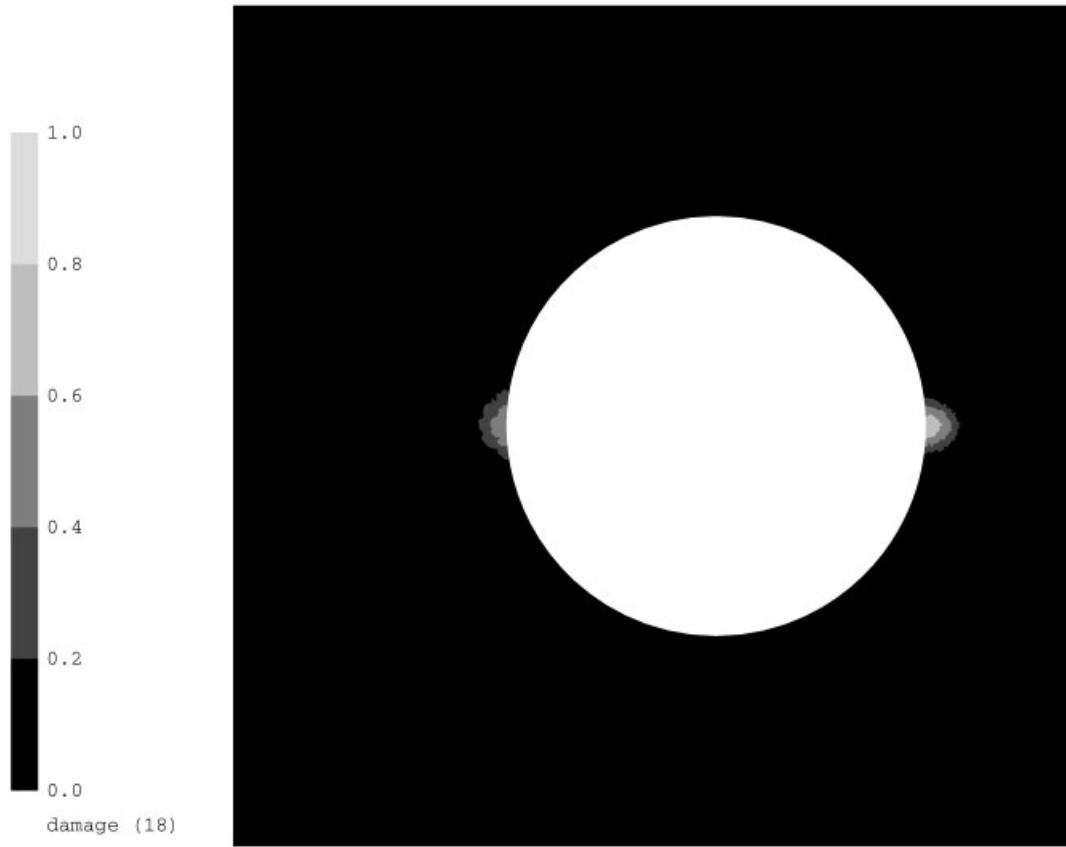


Figure 11. Damage field at the point **P**.

- to provide an example where the path-following constraint based on the maximal strain increment (10) is inefficient.

Even though the simulation is finally led on a real structure, the mechanical fields remain homogeneous, thanks to the non-local formulation, so that a closed-form solution can be derived.

6.1. Strain softening Drucker–Prager plasticity

A standard plasticity model with Drucker–Prager yield criterion [30] and linear isotropic strain softening is considered. The material state is defined through the strain $\boldsymbol{\varepsilon}$, the plastic strain $\boldsymbol{\varepsilon}^p$ and the cumulated plastic strain p . The stress–strain relation, the consistency condition and the flow rules are, respectively:

$$\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p = \frac{1 + \nu}{E} \boldsymbol{\sigma} - \frac{\nu}{E} (\text{tr } \boldsymbol{\sigma}) \mathbf{Id} \quad (43)$$

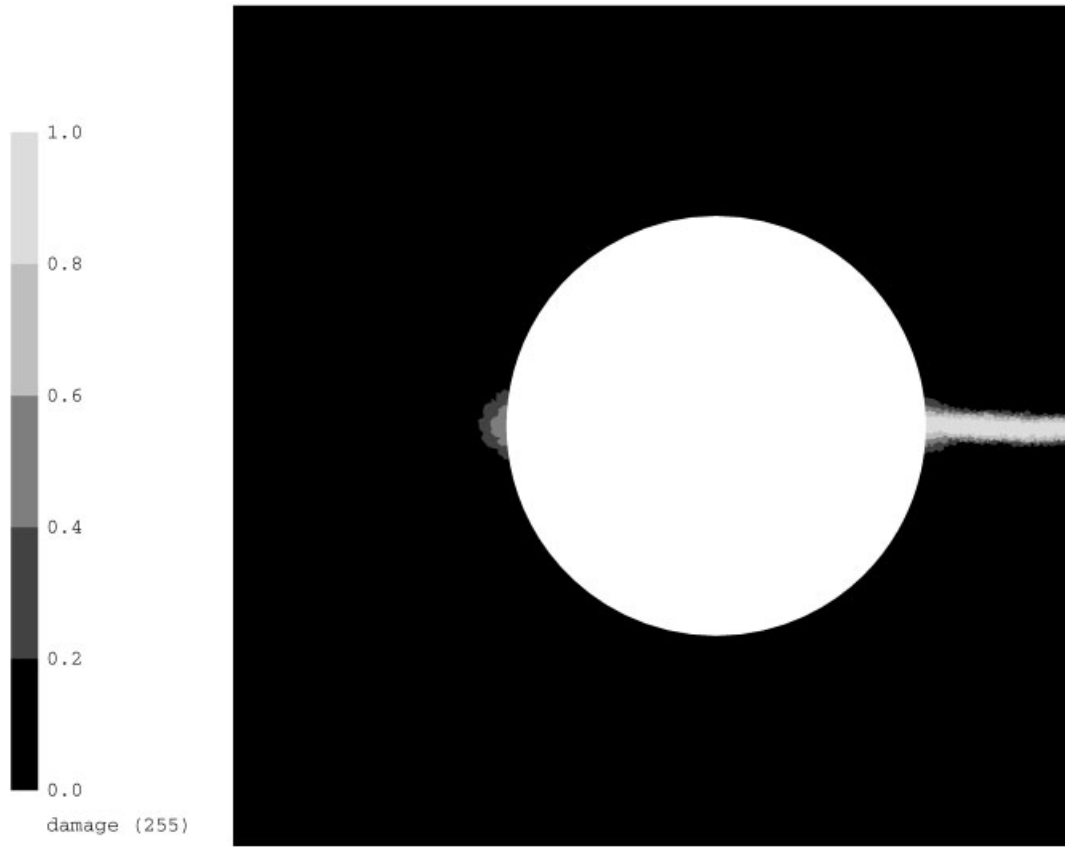


Figure 12. Damage field at point B.

$$f \leq 0 \quad \lambda \geq 0 \quad \lambda f \leq 0 \quad \text{where} \quad f(\boldsymbol{\sigma}, p) = \sigma_{\text{eq}} + \alpha \text{tr} \boldsymbol{\sigma} + hp - \sigma^y \quad (44)$$

$$\dot{\boldsymbol{\epsilon}}^p = \lambda \left(\frac{3}{2} \frac{\boldsymbol{\sigma}^D}{\sigma_{\text{eq}}} + \alpha \mathbf{Id} \right) \quad \text{and} \quad \dot{p} = \lambda \quad (45)$$

$\boldsymbol{\sigma}$ is the stress tensor, f the yield criterion, λ the plastic multiplier. E and ν are Young's modulus and Poisson's ratio which characterize the isotropic elastic behaviour. σ^y , $\alpha > 0$ and $h > 0$ are, respectively, the yield stress, the pressure dependency coefficient and the softening modulus; they characterize the plastic behaviour. Classical notations are used for the stress trace, the deviatoric stress and von Mises stress:

$$\text{tr} \boldsymbol{\sigma} = \sigma_{11} + \sigma_{22} + \sigma_{33} \quad \boldsymbol{\sigma}^D = \boldsymbol{\sigma} - \frac{1}{3}(\text{tr} \boldsymbol{\sigma})\mathbf{Id} \quad \sigma_{\text{eq}} = \sqrt{\frac{3}{2} \boldsymbol{\sigma}^D \cdot \boldsymbol{\sigma}^D} \quad (46)$$

It can be noticed that the elasticity domain is a cone in the stress-space and that softening results in its translation along the hydrostatic axis. A straightforward application of (13) shows

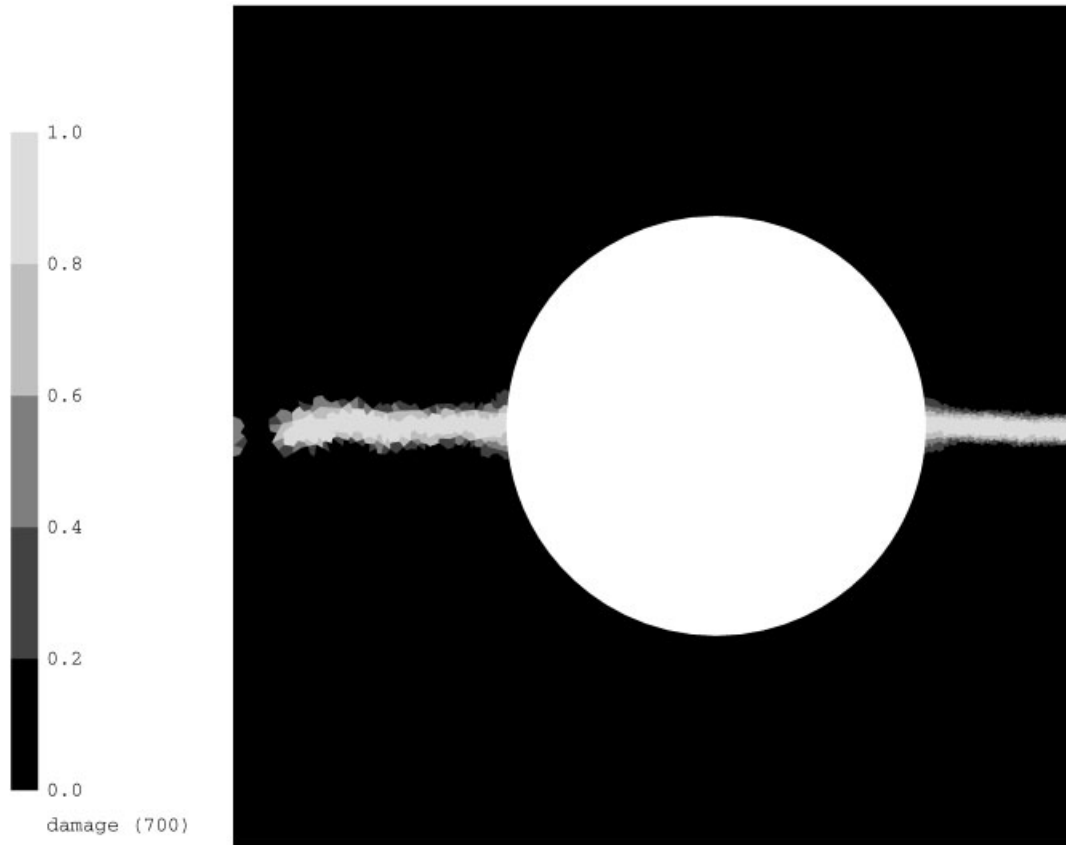


Figure 13. Damage field at point E.

that any strain path is licit if:[‡]

$$h < 3\mu + 9K\alpha^2 \quad \text{with} \quad 2\mu = \frac{E}{1+\nu} \quad \text{and} \quad 3K = \frac{E}{1-2\nu} \quad (47)$$

Once again, the elastic predictor is derived by setting the internal variables $\boldsymbol{\varepsilon}^{\mathbf{p}}$ and p to their values at the beginning of the increment $\boldsymbol{\varepsilon}^{\mathbf{p}^-}$ and p^- . According to (43) and (44), a normalized elastic predictor for the yield function is then

$$f^{\text{el}}(\boldsymbol{\varepsilon}) = \frac{1}{\sigma^y} [2\mu(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathbf{p}^-})_{\text{eq}} + 3K\alpha \text{tr}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathbf{p}^-}) + hp^- - \sigma^y] \quad (48)$$

[‡]Actually, because of the singular point on the conical yield surface, where another expression of the flow rule than (45) is required, the condition for any strain path to be licit is slightly more restrictive. However, this point has no influence on our application.

f^{el} is convex with respect to ε (sum of a semi-norm and a linear function), so that the scheme proposed in Section 3 is available. Thanks to the special form (48), the local equation (19) results again in a quadratic equation.

6.2. Snap-back in a simple compression test

Now, consider simple homogeneous tension and compression tests, say in direction 1. The equilibrium equations state that the only non-zero stress component is σ_{11} . Equations (43)–(45) reduce to (where the subscript 11 has been omitted and sgn denotes the sign function, which is equal to 1 for positive arguments and -1 otherwise):

$$\begin{aligned}\varepsilon - \varepsilon^{\text{p}} &= \frac{\sigma}{E} \\ f &\leq 0 \quad \lambda \geq 0 \quad \lambda f = 0 \quad \text{where} \quad f(\sigma, p) = |\sigma| + \alpha\sigma + hp - \sigma^y \\ \dot{\varepsilon}^{\text{p}} &= \lambda(\text{sgn } \sigma + \alpha) \quad \text{and} \quad \dot{p} = \lambda\end{aligned}\tag{49}$$

In tension, plasticity occurs when the stress σ reaches the tension yield stress σ^{t} , while in compression, it occurs when the stress σ reaches the compression yield stress $-\sigma^{\text{c}}$:

$$\sigma^{\text{t}} = \frac{\sigma^y}{1 + \alpha} \quad \text{and} \quad \sigma^{\text{c}} = \frac{\sigma^y}{1 - \alpha}\tag{50}$$

Focusing on compression, the stress magnitude $|\sigma|$ then decreases (softening) following a linear stress–strain curve:

$$\dot{\varepsilon} = \underbrace{\frac{\dot{\sigma}}{E}}_{\text{elastic unloading}} + \underbrace{\lambda\alpha}_{\text{plastic dilatancy}} - \underbrace{\lambda}_{\text{deviatoric plastic flow}} = \dot{\sigma} \left(\frac{1}{E} - \frac{(1 - \alpha)^2}{h} \right)\tag{51}$$

Consequently, there is a snap-back on the stress–strain curve in compression as soon as

$$h > E(1 - \alpha)^2\tag{52}$$

This snap-back originates from the cumulative effect of the elastic unloading and the plastic dilatancy which counterbalance the deviatoric plastic flow. Consequently, it is a purely local phenomenon (related to the behaviour) and not a structural effect, as in the two previous sections. It should be emphasized that such snap-backs are not related to an excessive simplicity of the model. They also occur with more complex laws, see for instance [31], even though the loading paths that exhibit snap-backs may be less intuitive than a simple compression test.

6.3. Numerical simulation

A simple compression test is simulated on a cubic specimen of edge length 100 mm and whose material parameters are given in Table III. They fulfil both condition (47) which ensures that any strain path is licit and condition (52) for the appearance of a snap-back. In addition to the compression loading, the sides of the cube are enforced to remain plane. To retrieve a well-posed problem, the nonlocal formulation described in Section 5 is applied again.

The computation is performed with a mesh composed of $10 \times 10 \times 10$ quadratic cubes (20 nodes for the displacements, 8 nodes for the smoothed strain and 8 Gauss points). Thanks to

Table III. Material parameters (simulation with Drucker–Prager non-local plasticity model).

Elasticity	Plasticity	Non-locality
$E = 20\,000\text{ MPa}$ $\nu = 0.3$	$\begin{cases} \sigma^t = 1.66\text{ MPa} \\ \sigma^c = 15\text{ MPa} \end{cases} \rightarrow \begin{cases} \sigma^y = 3\text{ MPa} \\ \alpha = 0.8 \end{cases}$ $h = 20\,000\text{ MPa}$	$l_c = 6\text{ mm}$

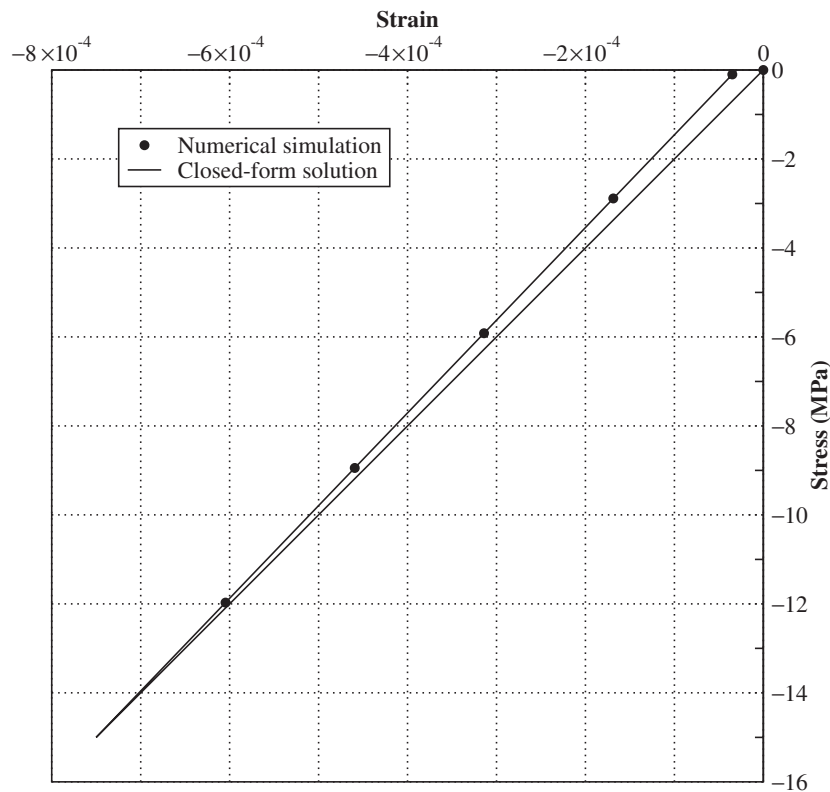


Figure 14. Drucker–Prager plasticity: stress–strain response.

the strain regularization, the mechanical fields remain homogeneous. The stress–strain response in the loading direction, plotted in Figure 14, is in total agreement with the closed-form solution given above and exhibits the expected snap-back. On the contrary, the path-following constraint based on the strain increment is not able to get over the snap-back since the strain paths turn back everywhere in the structure. This definitely proves the robustness of the MEP path-following constraint.

7. CONCLUSION

Three applications have showed in various contexts (cohesive zone model, non-local damage and non-local strain-softening plasticity) that the path-following constraint based on the maximal value of the elastic predictor for the yield function achieves the expected robustness, whereas the arc-length method as well as a path-following constraint based on the maximal strain increment may fail. It seems to confirm the importance of selecting only dissipative paths of solutions when sharp snap-backs occur. Besides, this robustness, which is in our opinion essential when dealing with real-life applications, does not require any prior knowledge of the structural behaviour, in particular where the dissipative processes take place. There is nevertheless a price to pay:

- a specific numerical solution of the local constraints has to be implemented for each constitutive law;
- the path-following constraint is a highly non-linear equation, but thanks to the solution algorithm proposed in Section 3, the cost of its solution has never exceeded 10% of the total computation time;
- the constitutive behaviour must rely on a threshold function to define an elastic predictor, a condition which precludes the use of the path-following constraint namely in the context of elastic buckling.

Despite this price, we think that the MEP path-following constraint is worth its cost. In addition, another advantage has not yet been mentioned: when the elastic predictor goes to zero, so does the incremental evolution of the internal variables, so that the time-continuous constitutive law is retrieved. This is generally not ensured by other path-following constraints. For instance, there is no continuous relation between the arc-length and the maximal increment of internal variables. Actually, we believe that the step-length $\Delta\tau$ may control the accuracy of the temporal discretization of the constitutive law; this aspect will be investigated in a future work.

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